

Doubt-to-Diagram — Pre-Hackathon Timeline (Frozen to Sept 20 MVP Submission)

Sept 2 – Sept 20 (18 days) → Idea/MVP Submission Deadline: Sept 20

This timeline is compressed from the original Sept 25 target to Sept 20, because Phase 1 requires submitting an idea package (title, description, video walkthrough, prototype URL, deck/document, proficiency fields, prior builds, standout pitch) ahead of the actual Hyderabad battle.

PART 1: Research, Setup & Architecture (Sept 2–6)

Goal: Lock the tech stack, set up environments, and confirm feasibility — no submission-facing work yet.

Flow

1. Finalize scope — pick ONE guaranteed subject for the MVP demo (e.g., physics kinematics); math/chemistry as stretch.
2. Choose models: OCR engine, VLM (Phi-3-vision or alt), LLM (Gemma 2B/Phi-3), optional Whisper Tiny + TTS.
3. Set up Pixel 6a as test device (developer mode, runtime installed — MediaPipe/TFLite/ONNX Mobile).
4. Create GitHub repo with base project structure; grant teammate access.
5. Confirm team roles, comms channel, and who is team leader (only the leader can submit the idea form).
6. Draft a rough one-paragraph problem/solution summary — early input for the idea title and description.

Deliverables by Sept 6

- ☐ Tech stack finalized and documented
- ☐ Pixel 6a environment ready
- ☐ GitHub repo live
- ☐ Team leader + roles confirmed
- ☐ Rough problem/solution paragraph drafted

PART 2: Core Pipeline Build (Sept 7–11)

Goal: Get camera → OCR → model inference → text output working end-to-end on Pixel 6a. This becomes the proof behind your prototype URL.

Flow

1. Build camera capture UI (button, live preview, image confirmation).
2. Integrate OCR on captured image; test against sample handwritten question images.
3. Add image pre-processing step (contrast/low-light cleanup, adapted from your prior enhancement work).
4. Load and run quantized VLM/LLM on OCR output; generate raw text explanation.
5. Benchmark latency per step, peak RAM, thermal behavior on repeated runs.
6. Identify a fallback smaller/quantized model in case of latency issues on iQOO hardware.
7. Push working build to GitHub — this becomes your **prototype URL** for the submission form.

Deliverables by Sept 11

- [] Camera → OCR → model → text output working on Pixel 6a
- [] Latency/RAM/thermal benchmarks logged
- [] Fallback model identified
- [] GitHub repo updated and public/shareable (prototype URL ready)

PART 3: Feature Completion & Integration (Sept 12–16)

Goal: Add remaining features and integrate into one coherent flow — this becomes the substance of your description and video walkthrough.

Flow

1. Add regional language support (Telugu/Hindi/English prompt templates); test output quality.
2. Build diagram generation module (template-based renderer for common diagrams).
3. Integrate Whisper Tiny + TTS for voice narration (optional, adds "creative phone use" point).
4. Wire all modules into one app flow: capture → OCR → pre-process → model → explanation → diagram → voice.
5. Add basic error handling (blurry image, unparseable question, model timeout fallback).
6. Run one full Red-Light-style rehearsal on Pixel 6a: phone only, no laptop, timed 2–3 hours.
7. Record a short screen-capture video of the working flow — draft cut for your **video walkthrough URL**.

Deliverables by Sept 16

- [] Regional language output working
- [] Diagram generation working for at least one subject
- [] Full app flow integrated end-to-end
- [] Error handling in place
- [] Rehearsal completed with notes
- [] Draft walkthrough video recorded

PART 4: MVP Freeze & Submission Package (Sept 17–20)

Goal: Stop building new features. Polish what exists and assemble the submission form completely and convincingly.

Flow

1. **Feature freeze on Sept 17** — fix only bugs from here, no new features.
2. Finalize the working MVP build; confirm it runs reliably at least 5 times in a row without crashing.
3. Edit the walkthrough video to 60–90 seconds: show camera capture → explanation → diagram output clearly, upload to YouTube (unlisted is fine) for the **video walkthrough URL**.
4. Confirm the **prototype URL**: either a live demo link or the GitHub repo with clear README and setup steps.
5. Build the **deck/document** (PDF/PPT, under 25MB): problem, solution, architecture diagram, tech stack, demo screenshots, impact, team.
6. Write the **idea title** — short and punchy (e.g., "Doubt-to-Diagram: Offline AI Tutor in Your Pocket").
7. Write the **description** (50+ words minimum): what you're building, for whom, and how — cover the offline, regional-language, on-device NPU angle explicitly.
8. Fill **Android proficiency** and **LLM proficiency** fields honestly but favorably — list specific frameworks/models you used (TFLite, MediaPipe, Gemma 2B, Phi-3, etc.).
9. Fill **prior builds & hackathons** — mention your bird-sound classification (CNN/EfficientNet), low-light image enhancement research, and any prior hackathon/internship work.
10. Write "**what makes you stand out**" — emphasize domain edge: CV + audio ML background, prior RAG/education product experience (Learnee), and why an offline regional-language AI tutor matters for underserved students.
11. Team leader reviews the full form, then submits before the Sept 20 deadline.

Deliverables by Sept 20 (Submission Day)

- ☐ MVP build stable and demo-ready
- ☐ Video walkthrough (60–90 sec) uploaded and linked
- ☐ Prototype URL live and accessible
- ☐ Deck/document finalized and uploaded (PDF/PPT under 25MB)
- ☐ Idea title finalized
- ☐ Description written (50+ words, covers what/for whom/how)
- ☐ Android + LLM proficiency fields filled
- ☐ Prior builds & hackathons section filled
- ☐ "What makes you stand out" section filled
- ☐ Form submitted by team leader before deadline

After Submission → Hyderabad Battle (Sept 26–27)

Once the idea is submitted, remaining time before check-in should go toward:

- Rehearsing the live demo further
- Practicing Office Kit pairing logistics (only testable on real iQOO hardware at check-in)
- Preparing a backup video in case live inference is slow on the actual device

Scoring Weights to Keep in Mind Throughout

- Device data (phone + Office Kit usage): 25%
- Product quality: 30%
- Novelty/impact: 20%
- Technical depth: 15%
- Demo/pitch: 10%

For the Sept 20 submission specifically, product quality and novelty/impact matter most — reviewers are reading your description, deck, and prototype link before they ever see the phone-based scoring.